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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M.Tech (Construction Technology) (CBCS) Degree Examinations

May - June 2024

22CCT212 – INFRASTRUCTURE DEVELOPMENT – PROGRAMMES, PLANNING AND APPRAISAL

Duration: 3 Hours

Max. Marks:100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and one full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1.	a) How do the infrastructure development policies of central and state governments in India align with broader economic development goals?	10	L*3	1	1,3
	b) Evaluate the economic, social, and environmental impacts of major infrastructure projects such as the construction of new highways or the expansion of urban metro systems. How do these projects contribute to economic growth, social equity, and sustainability?	10	L3	2	1,5,6
2.	a) Propose a comprehensive framework for integrating various infrastructure development initiatives across different sectors in India. How can synergy between road, rail, air, and urban infrastructure projects be achieved?	10	L3	1	1,6,7
	b) Propose innovative financing mechanisms and public-private partnerships (PPPs) to fund large-scale infrastructure projects in India. How can these approaches attract private investment while ensuring public accountability and value for money?	10	L3	2	1,5,6
3.	a) Evaluate the effectiveness of past infrastructure development programs in India, such as the Bharatmala, Sagarmala, and Smart Cities Mission, in achieving their stated objectives. What lessons can be learned from these initiatives for future planning?	10	L3	1	1,3
	b) Compare and contrast the approaches to infrastructure planning and development between different regions in India. What factors contribute to variations in infrastructure quality and accessibility across states?	10	L3	2	1,5,6
Unit – II					
4.	a) How would you evaluate the effectiveness of project-wise studies in generating and assessing alternatives for project development?	10	L3	3	1,3,5
	b) Analyze the process of developing alternatives for a project. What criteria should be considered when evaluating different alternatives, and how can stakeholders contribute to this process?	10	L3	4	1,5,6
5.	a) Evaluate the effectiveness of the introduction and screening phase in project planning. How do these phases contribute to the selection of viable project options?	10	L3	3	1,5,6
	b) Assess the significance of various planning tools commonly used in project management. How do these tools aid project managers in optimizing resources, timelines, and budgets?	10	L3	4	1,7,8
6.	a) Evaluate the role of financial analysis in project appraisal. How does financial analysis assist stakeholders in evaluating the economic viability and return on investment of a project?	10	L3	3	1,5,6
	b) Analyze the importance of economic analysis in project appraisal. How does economic analysis help in assessing the broader economic implications and benefits of a project?	10	L3	4	1,7,8

P.T.O

Unit – III

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| 7. a) | How can sustainable infrastructure development be leveraged to address the interconnected challenges of climate change, urbanization, and resource scarcity? | 10 | L3 | 5 | 1,6,7 |
| b) | Analyze the effectiveness of risk assessment methodologies in identifying and mitigating potential risks in infrastructure projects. What strategies can be employed to enhance the robustness and accuracy of risk assessments? | 10 | L3 | 5 | 1,6,7 |
| 8. a) | Analyze the economic implications of integrating sustainability principles into infrastructure projects. What are the long-term cost savings and benefits associated with sustainable infrastructure compared to traditional approaches? | 10 | L3 | 5 | 1,7,8 |
| b) | Critically assess the role of uncertainty in infrastructure development projects. How can stakeholders proactively manage and respond to uncertainties to minimize project disruptions and delays? | 10 | L3 | 5 | 1,7,8 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
